

TrES-5b

R-0800

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_220417 · created 2026-07-28T09:08:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459686.9261 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.05
Transit depth (Rp/Rs)^2	0.93 +/- 0.97 [%]
Orbital Inclination (inc)	87.25 +/- 3.08
Airmass coefficient 1 (a1)	0.952 +/- 0.044
Airmass coefficient 2 (a2)	0.044 +/- 0.061
Scatter in the residuals of the lightcurve fit is	4.59 %
Best Comparison Star	#2 - [607, 297]
Optimal Aperture	2.44
Optimal Annulus	10.45
Transit Duration (day)	0.0657 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.096 $\pm$ 0.05 vs 0.142 (deviation 0.92 σ)
Duration fit vs published	0.0657 d vs 0.0758 d (deviation 1.04 σ)
Mid-time O-C	1.2 min
Depth SNR	1.9
Residual scatter	4.59 %

Light curve

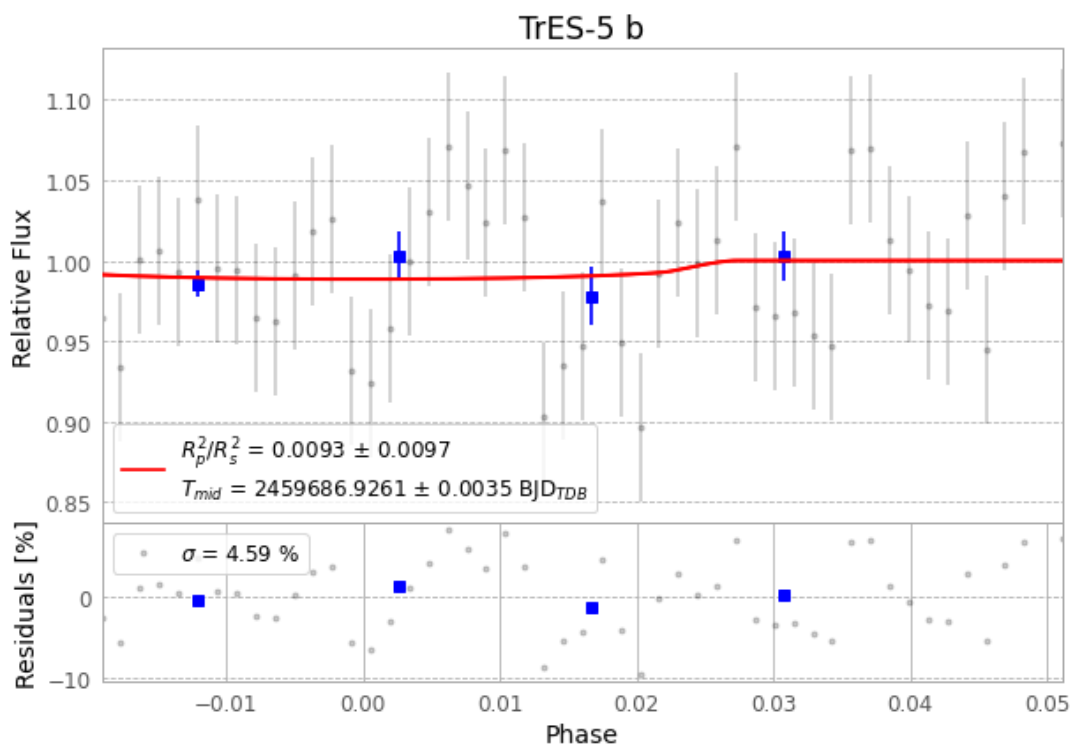


Figure 1 — FinalLightCurve_TrES-5 b_2022-04-17.png (SHA-256 cd0c727ccd9cd5af...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 341], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4e3b03a3a79ade6e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

54 FITS frames (35 MB), TRES-5220417093315.FITS → TRES-5220417122712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-220417.log (c626b13c806b...), FinalLightCurve_TrES-5 b_2022-04-17.png (cd0c727ccd9c...), FinalLightCurve_TrES-5 b_2022-04-17.csv (fe24416fd23c...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 09:08Z.